

BABU BANARASI DAS
UNIVERSITY
SCHOOL OF COMPUTER APPLICATION



BIG DATA FUNDAMENTAL

GROUP PROJECT:

- **FOOD ORDERS DATA ANALYSIS - A CASE STUDY**

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Submitted to:

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Section:

BCADS25

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PROJECT

GROUP D

Food Orders Dataset MapReduce Practical

➤ Architecture:

This project follows a Big Data pipeline using Hadoop ecosystem.

➤ Data Flow:

**Raw Dataset → HDFS → MapReduce Processing
→ Processed Data → Hive → Insights**

➤ Explanation:

- Dataset is stored in HDFS**
- MapReduce processes the data**
- Processed data is analyzed using Hive**
- Insights are generated**

➤ Tools Used:

- **HDFS**
- **MapReduce**
- **Hive**
- **Python**
- **Google Colab**

➤ Technologies:

- **CSV Dataset**
- **Hadoop Ecosystem**
- **Python Programming**

Use of Generative AI in the Project

AI tools were used to assist in development and understanding.

➤ Areas Where AI Was Used:

1. **Code Generation**
2. **Debugging**
3. **Query Writing**
4. **Concept Understanding**

➤ Benefits:

- **Faster development**
- **Reduced errors**
- **Better understanding**
- **Improved efficiency**

➤ Validation:

All AI-generated content was reviewed and verified.

Dataset Description

The dataset used is a food orders dataset in CSV format.

➤ Columns:

- **item_name**
- **quantity**
- **item_price**
- **order_id**
- **Location**

➤ Purpose:

- **Analyze food order patterns**
- **Calculate total orders**
- **Generate insights**

➤ Dataset Source:

Online dataset for project use.

Conclusion

This project demonstrates the use of Big Data tools to analyze food order data.

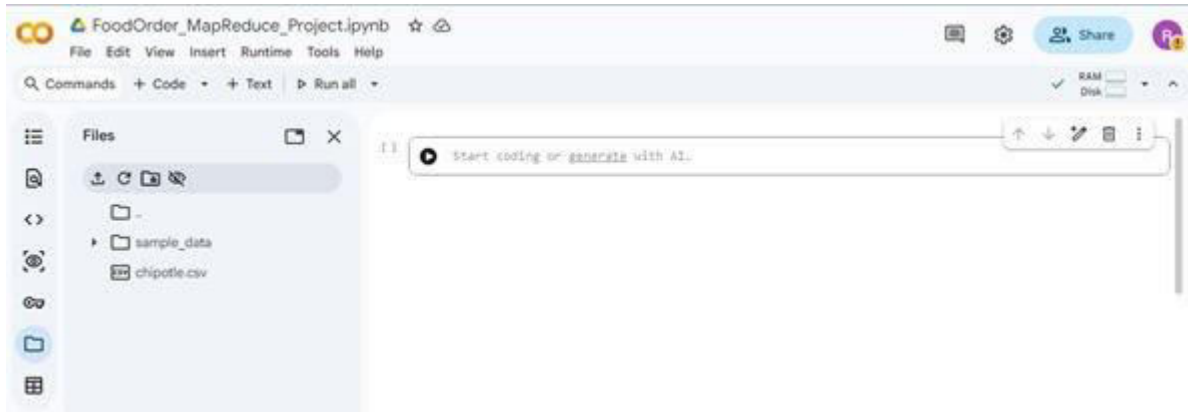
- **Data was stored in HDFS**
- **Processed using MapReduce**
- **Analyzed using Hive**

The project helped in understanding Big Data workflow and data analysis techniques.

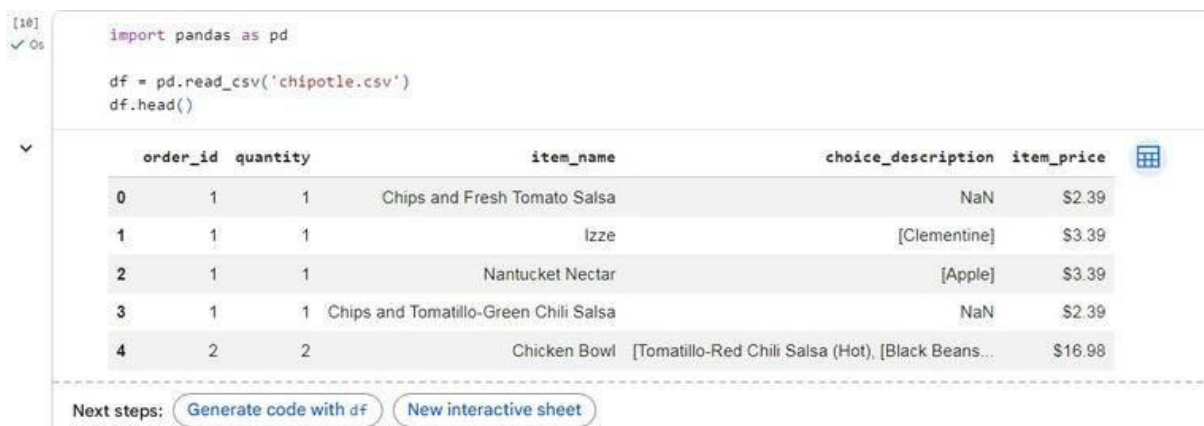
AI tools improved efficiency and learning.

MapReduce Implementation Tasks

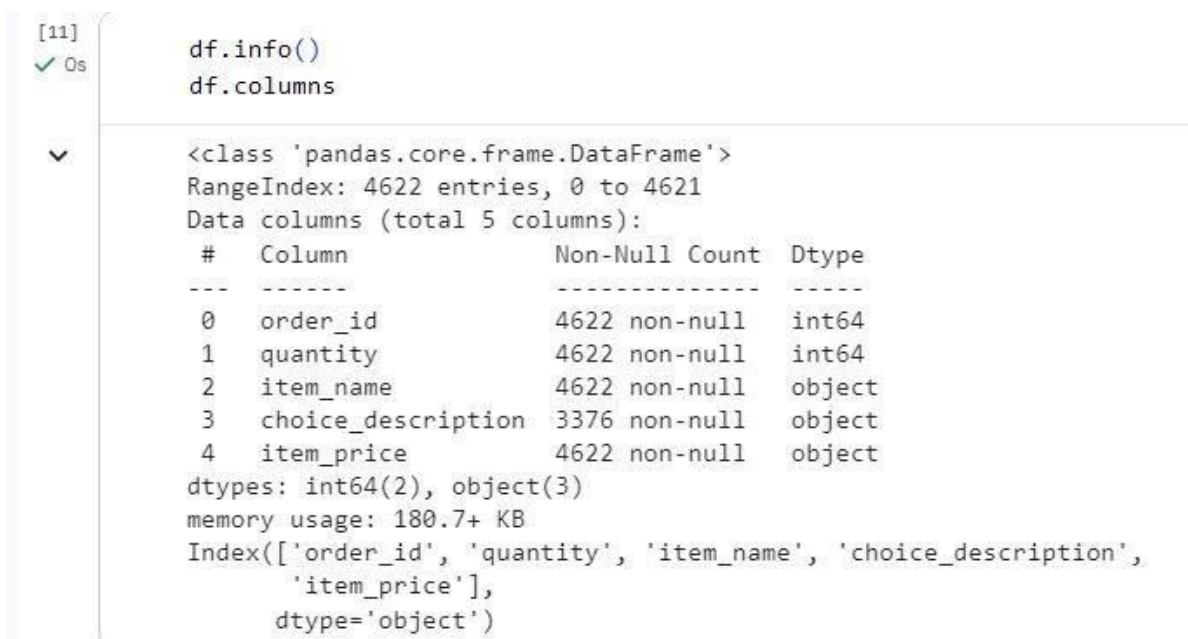
1. Upload chipotle.csv in Google Collab under <Files> section



2. Dataset load code



3. Data info check



4. Data Cleaning (Price Column)

```
[12]
✓ Os
df['item_price'] = df['item_price'].str.replace('$', '')
df['item_price'] = df['item_price'].astype(float)
```

5. Mapper Function

```
[13]
✓ Os
def mapper(df):
    mapped = []
    for index, row in df.iterrows():
        mapped.append((row['item_name'], row['quantity']))
    return mapped

mapped_data = mapper(df)

mapped_data[:10]

[('Chips and Fresh Tomato Salsa', 1),
 ('Izze', 1),
 ('Nantucket Nectar', 1),
 ('Chips and Tomatillo-Green Chili Salsa', 1),
 ('Chicken Bowl', 2),
 ('Chicken Bowl', 1),
 ('Side of Chips', 1),
 ('Steak Burrito', 1),
 ('Steak Soft Tacos', 1),
 ('Steak Burrito', 1)]
```

6. Run Mapper

```
[28]
✓ Os
▶ mapped_data = mapper(df)
  mapped_data[:10]

... [('Chips and Fresh Tomato Salsa', 1),
      ('Izze', 1),
      ('Nantucket Nectar', 1),
      ('Chips and Tomatillo-Green Chili Salsa', 1),
      ('Chicken Bowl', 2),
      ('Chicken Bowl', 1),
      ('Side of Chips', 1),
      ('Steak Burrito', 1),
      ('Steak Soft Tacos', 1),
      ('Steak Burrito', 1)]
```

7. Shuffle and Sort

```
[30]
✓ 0s  from collections import defaultdict

shuffled = defaultdict(list)

for key, value in mapped_data:
    shuffled[key].append(value)

list(shuffled.items())[:5]
```

▼ ...

1,
1,
1,
1,
1,
1,
1,
1,
1,
1,
1,
1,
1,
1,
1,
1,

8. Display Shuffle Output

```
[31]
✓ 0s  for key, value in list(shuffled.items())[:5]:
    print(key, value[:10])
```

▼ ...

Chips and Fresh Tomato Salsa [1, 1, 1, 1, 1, 1, 1, 1, 1, 1]
Izze [1, 1, 1, 1, 1, 1, 1, 1, 1, 1]
Nantucket Nectar [1, 1, 1, 1, 1, 1, 2, 1, 1, 1]
Chips and Tomatillo-Green Chili Salsa [1, 1, 1, 1, 1, 1, 1, 1, 1, 1]
Chicken Bowl [2, 1, 1, 1, 1, 1, 1, 1, 1, 1]

9. Reducer Function and Run Reducer

```
[32] def reducer(shuffled):  
    ✓ Os    reduced = {}  
           for key in shuffled:  
               reduced[key] = sum(shuffled[key])  
           return reduced  
  
    result = reducer(shuffled)  
  
    # Top 10 results dekhne ke liye  
    list(result.items())[:10]
```

...

```
[('Chips and Fresh Tomato Salsa', 130),  
 ('Izze', 20),  
 ('Nantucket Nectar', 29),  
 ('Chips and Tomatillo-Green Chili Salsa', 33),  
 ('Chicken Bowl', 761),  
 ('Side of Chips', 110),  
 ('Steak Burrito', 386),  
 ('Steak Soft Tacos', 56),  
 ('Chips and Guacamole', 506),  
 ('Chicken Crispy Tacos', 50)]
```

10. Convert to DataFrame

```
[33] import pandas as pd  
    ✓ Os  
  
    result_df = pd.DataFrame(list(result.items()), columns=['Food_Item', 'Total_Quantity'])  
  
    result_df.head()
```

...

	Food_Item	Total_Quantity
0	Chips and Fresh Tomato Salsa	130
1	Izze	20
2	Nantucket Nectar	29
3	Chips and Tomatillo-Green Chili Salsa	33
4	Chicken Bowl	761

Next steps: [Generate code with result_df](#) [New interactive sheet](#)

11. Sort Data

```
[34] result_df = result_df.sort_values(by='Total_Quantity', ascending=False)
result_df.head(10)
```

	Food_Item	Total_Quantity
4	Chicken Bowl	761
11	Chicken Burrito	591
8	Chips and Guacamole	506
6	Steak Burrito	386
33	Canned Soft Drink	351
19	Chips	230
21	Steak Bowl	221
16	Bottled Water	211
0	Chips and Fresh Tomato Salsa	130
12	Canned Soda	126

12. Top 5 items

```
[35] top_items = result_df.head(5)
top_items
```

	Food_Item	Total_Quantity
4	Chicken Bowl	761
11	Chicken Burrito	591
8	Chips and Guacamole	506
6	Steak Burrito	386
33	Canned Soft Drink	351

Next steps: [Generate code with top_items](#) [New interactive sheet](#)

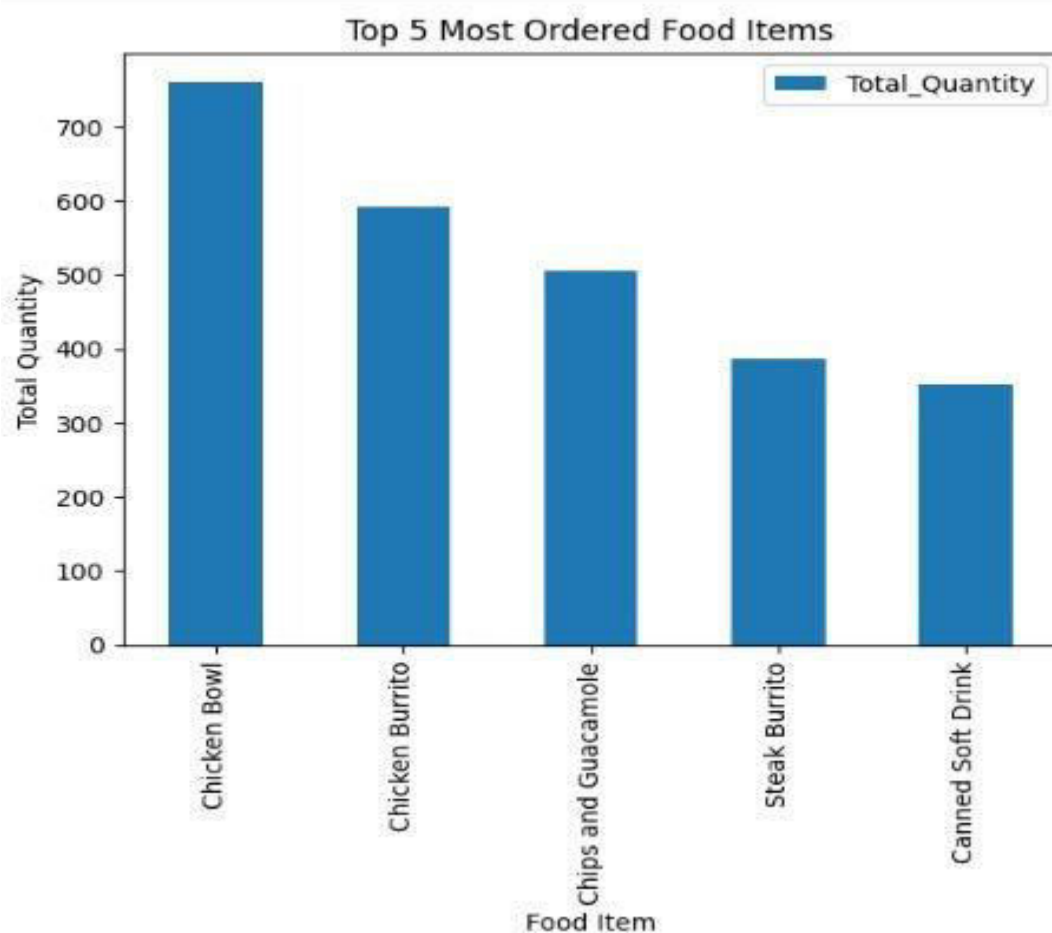
13. Graph Plot

```
[36] import matplotlib.pyplot as plt

top_items.plot(x='Food_Item', y='Total_Quantity', kind='bar')

plt.title("Top 5 Most Ordered Food Items")
plt.xlabel("Food Item")
plt.ylabel("Total Quantity")

plt.show()
```



14. Revenue Calculation

```
[37]: df['revenue'] = df['quantity'] * df['item_price']  
✓ Os revenue = df.groupby('item_name')['revenue'].sum().sort_values(ascending=False)  
revenue.head(5)
```

```
... revenue  
item_name  
Chicken Bowl 8044.63  
Chicken Burrito 6387.06  
Steak Burrito 4236.13  
Steak Bowl 2479.81  
Chips and Guacamole 2475.62  
dtype: float64
```

HDFS Implementation Tasks

1) Create directories in HDFS

```
hdfs dfs -mkdir /input  
hdfs dfs -mkdir /output
```

```
(base) rashirai@Rashis-MacBook-Air ~ % mkdir -p ~/hdfs/input  
[mkdir -p ~/hdfs/output
```

2) Upload dataset to HDFS (/input)

```
hdfs dfs -put chipotle_dataset.csv /input/
```

```
[(base) rashirai@Rashis-MacBook-Air ~ % cp ~/Downloads/chipotle_dataset.csv ~/hdfs/input/  
[(base) rashirai@Rashis-MacBook-Air ~ % ls ~/hdfs/input
```

3) Verify storage (list contents)

```
hdfs dfs -ls /input
```

```
[(base) rashirai@Rashis-MacBook-Air ~ % cp ~/Downloads/chipotle_dataset.csv ~/hdfs/input/  
[(base) rashirai@Rashis-MacBook-Air ~ % ls ~/hdfs/input  
chipotle_dataset.csv
```

4) View dataset (verify content)

```
hdfs dfs -cat /input/chipotle_dataset.csv
```

```

order_id,quantity,item_name,choice_description,item_price
1,1,Chips and Fresh Tomato Salsa,NULL,2.39
1,1,Izze,[Clementine],3.39
1,1,Nantucket Nectar,[Apple],3.39
1,1,Chips and Tomatillo-Green Chili Salsa,NULL,2.39
2,2,Chicken Bowl,[Tomatillo-Red Chili Salsa (Hot), [Black Beans, Rice, Cheese, Sour Cream]],16.98
3,1,Chicken Bowl,[Fresh Tomato Salsa (Mild), [Rice, Cheese, Sour Cream, Guacamole, Lettuce]],10.98
3,1,Side of Chips,NULL,1.69
4,1,Steak Burrito,[Tomatillo Red Chili Salsa, [Fajita Vegetables, Black Beans, Pinto Beans, Cheese, Sour Cream, Guacamole, Lettuce]],11.75
4,1,Steak Soft Tacos,[Tomatillo Green Chili Salsa, [Pinto Beans, Cheese, Sour Cream, Lettuce]],9.25

```

5) Store processed output in /output

(Example: copy or move result file)

hdfs dfs -cp /input/chipotle_dataset.csv /output/

```
(base) rashirai@Rashis-MacBook-Air ~ % hdfs dfs -cp /input/chipotle_dataset.csv /output/
```

6) Verify output directory

hdfs dfs -ls /output

```
(base) rashirai@Rashis-MacBook-Air ~ % ls ~/hdfs/output
sample_output.csv
```

HIVE IMPLEMENTATION

• Uploading Dataset on Google colab

```
from google.colab import files  
uploaded = files.upload()
```

```
[2] ✓ 16s  
▶ from google.colab import files  
  uploaded = files.upload()  
  
▼ ... Choose files chipotle_dataset.csv  
  chipotle_dataset.csv(text/csv) - 360352 bytes, last modified: 25/04/2026 - 100% done  
  Saving chipotle_dataset.csv to chipotle_dataset.csv
```

Reading the file

```
import pandas as pd  
df = pd.read_csv('chipotle_dataset (1).csv', on_bad_lines='skip')  
df.head()
```

```
▶ import pandas as pd  
  
df = pd.read_csv('chipotle_dataset (1).csv', on_bad_lines='skip')  
df.head()
```

...	order_id	quantity	item_name	choice_description	item_price	
0	1	1	Chips and Fresh Tomato Salsa	NaN	2.39	
1	1	1	Izze	[Clementine]	3.39	
2	1	1	Nantucket Nectar	[Apple]	3.39	
3	1	1	Chips and Tomatillo-Green Chili Salsa	NaN	2.39	
4	3	1	Side of Chips	NaN	1.69	

Next steps: [Generate code with df](#) [New interactive sheet](#)

✓ Group By

```
df.groupby("item_name")["quantity"].sum()
```

[6]
✓ 0s

```
df.groupby("item_name")["quantity"].sum()
```

...	quantity
item_name	
6 Pack Soft Drink	55
Barbacoa Bowl	5
Barbacoa Burrito	3
Barbacoa Salad Bowl	2
Bottled Water	211
Canned Soda	126
Canned Soft Drink	351
Carnitas Burrito	1
Carnitas Crispy Tacos	1
Carnitas Soft Tacos	3
Chicken Bowl	7
Chicken Burrito	7
Chicken Crispy Tacos	2
Chicken Salad Bowl	2
Chicken Soft Tacos	1
Chips	230
Chips and Fresh Tomato Salsa	130

	Chips and Guacamole	506
...	Chips and Mild Fresh Tomato Salsa	1
	Chips and Roasted Chili Corn Salsa	23
	Chips and Roasted Chili-Corn Salsa	18
	Chips and Tomatillo Green Chili Salsa	45
	Chips and Tomatillo Red Chili Salsa	50
	Chips and Tomatillo-Green Chili Salsa	33
	Chips and Tomatillo-Red Chili Salsa	25
	Crispy Tacos	2
	Izze	20
	Nantucket Nectar	29
	Side of Chips	110
	Steak Bowl	2
	Steak Burrito	5
	Steak Crispy Tacos	1
	Steak Soft Tacos	2
	Veggie Burrito	1

dtype: int64

✓ ORDER BY

```
df.sort_values(by="item_price", ascending=False)
```

[7] ✓ Os

```
df.sort_values(by="item_price", ascending=False)
```

	order_id	quantity	item_name	choice_description	item_price
1375	1443	15	Chips and Fresh Tomato Salsa	NaN	44.25
1388	1454	3	Chicken Burrito	[Fresh Tomato Salsa]	26.25
1732	1786	4	Chips and Guacamole	NaN	17.80
344	366	2	Chicken Crispy Tacos	[Tomatillo Red Chili Salsa]	17.50
162	181	2	Chicken Bowl	[Tomatillo Red Chili Salsa]	17.50
...	...	...	...	...	...
1411	1484	1	Canned Soda	[Dr. Pepper]	1.09
1412	1484	1	Canned Soda	[Dr. Pepper]	1.09
1413	1485	1	Canned Soda	[Diet Dr. Pepper]	1.09
35	38	1	Bottled Water	NaN	1.09
42	47	1	Canned Soda	[Dr. Pepper]	1.09

1795 rows x 5 columns

✓ TOP ITEMS

```
df.groupby("item_name")["quantity"].sum().sort_values(ascending=False).head()
```

[8] ✓ Os

```
df.groupby("item_name")["quantity"].sum().sort_values(ascending=False).head()
```

	quantity
Chips and Guacamole	506
Canned Soft Drink	351
Chips	230
Bottled Water	211
Chips and Fresh Tomato Salsa	130

dtype: int64

ACKNOWLEDGEMENT

We would like to express our sincere gratitude to our respected teacher, Mr. Vikas (IBM), for his valuable guidance, support, and encouragement throughout the completion of this project.

This project has been a great learning experience for us, helping us understand the practical implementation of Big Data technologies such as HDFS, MapReduce, and Hive.

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Lastly, we acknowledge the use of Generative AI tools that assisted us in understanding concepts and improving the overall quality of this project.